

GENERAL SPECIFICATION G1.4

Metalwork

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G1.4 METALWORK

G1.4.1 General

- (a) This section covers the requirements for the structural metalwork, ladders, hand railings, platforms (permanent and mobile) and others, to designs by the Contractor where such designs are proposed as part of the Permanent Works. This part applies to all metal work including work in steel, stainless steel and aluminium.
- (b) Contractor's Design Responsibility: Except where the necessary details are shown on the S.O.'s drawings, the Contractor shall carry out the detailed design of connections between members (both bolted and welded) to the design data approved or provided by the S.O..

G1.4.2 Reference Standards

- (a) Unless otherwise specified, structural metalwork shall comply with the relevant Singapore Standards and the Reference Standards listed below, where applicable:

Standard	Subject
BS EN 10034	Structural steel I and H sections. Tolerances on shape and dimensions.
BS EN ISO 9606-1	Qualification testing of welders. Fusion welding. Steels.
BS EN ISO 15607	Specification and qualification of welding procedures for metallic materials. General rules.
BS EN ISO 2560	Welding consumables. Covered electrodes for manual metal arc welding of non-alloy and fine grain steels. Classification.
BS EN 573	Wrought aluminium and aluminium alloys - plate, sheet and strip.
BS EN 754	Wrought aluminium and aluminium alloys - drawn tube.
BS EN 603	Wrought aluminium and aluminium alloys for general engineering purposes - forging stock and forging.
BS EN 755	Wrought aluminium and aluminium alloys - bars, extruded round tubes and sections.
BS EN ISO 17632	Welding consumables. Tubular cored electrodes for gas shielded and non-gas shielded metal arc welding of non-alloy and fine grain steels. Classification.
BS EN 1011	Arc welding of carbon and carbon manganese steels.
BS EN 1993-6	The design and testing of steel overhead runway beams.
BS 3692	ISO metric precision hexagon bolts, screws and nuts. Metric units.
BS 4190	ISO metric black hexagon bolts, screws and nuts.
BS 4320	Metal washers for general engineering purposes. Metric series.

Standard	Subject
BS EN 14399-1 to 10	High-strength structural bolting assemblies for preloading. General requirements.
BS EN 1993-1-8	Eurocode 3. Design of steel structures. Design of joints.
PD 5500	Specification for Unfired fusion welded pressure vessels.
BS EN 1993-1-10	Eurocode 3. Design of steel structures. Material toughness and through-thickness properties.
BS EN 1999-1-1:2007+A2	Eurocode 9: Design of aluminium structures. General structural rules.
BS EN 10034	Structural steel I and H sections.
BS EN 10025	Hot rolled products of structural steels. General technical delivery conditions.
BS EN 14399-1	High strength friction grip bolts.
BS EN 1993-1-10	Eurocode 3. Design of steel structures. Material toughness and through-thickness properties.
BS EN 1999-1-1:2007+A2	Eurocode 9: Design of aluminium structures. General structural rules.
BS EN 10029	Hot-rolled steel plates 3 mm thick or above. Tolerances on dimensions and shape.
BS EN 10210	Hot finished structural hollow sections of non-alloy and fine grain steels.
BS EN 10056, BS EN 10067, BS EN 10210	Hot-rolled structural steel sections.
BS EN ISO 14122 Parts 1 to 3	Safety of machinery. Permanent means of access to machinery.

G1.4.3 Submissions by the Contractor

- (a) The following submissions are required:
- (i) Professional Engineer (Singapore) endorsed drawings required for submission or as record of submissions to be made by the Contractor to statutory authorities as part of the Works.
 - (ii) Drawings comprising general arrangements, general assembly, detailed shop and erection including identification of members.
 - (iii) Professional Engineer (Singapore) endorsed data: Calculations for structural analysis and design where this is carried out by the Contractor, calculations for connection details, where requested by the S.O. and welding procedure sheets. Manufacturers' catalogues (where alternatives are offered for approval), Manufacturer's catalogues and test data for proprietary fixing bolts.

- (iv) Quality Control Submittals:
 - 1) Material Certificates;
 - 2) Laboratory Testing Equipment or Instruments: Certified calibrations, manufacturer's product data, and test procedures;
 - 3) Certified welding inspection and test results;
 - 4) Weld Inspection and Testing Agency: Certification and qualifications;
 - 5) Welding Inspector: Certification and qualifications;
 - 6) List of qualified welders and welding operators;
 - 7) Current test records for qualified welder(s) or welding operator(s) and weld type(s) for factory and field welding (WPQ);
 - 8) Weld Procedures: Records in accordance with ASME Boiler and Pressure Vessel Code, Section IX for weld type(s) and base metal(s);
 - 9) Shop and field welding procedure specifications (WPS);
 - 10) Procedure qualification records (PQR);
 - 11) Testing agency personnel credentials;
 - 12) Non-destructive inspection and testing procedures; and
 - 13) Manufacturer's Certification of Compliance for welding electrodes and filler metal.
- (v) Any other inspection certificates and qualification certificates as requested by S.O..

G1.4.4 Materials

- (a) General
 - (i) The Contractor shall provide items of metalwork shown on the Drawings or specified herein, together with all fixings, mounting brackets, locking screws, lifting keys and other accessories normally required for erection, permanent fixing and operation.
 - (ii) The Contractor shall ensure that the supplied material is suitable for the environment/area of application and of uniform design throughout the plant.
 - (iii) No orders for engineering metalwork shall be placed by the Contractor until he has submitted his detailed drawings, and where applicable supporting design calculations, to the S.O. and received the S.O.'s approval.
 - (iv) Where the Contractor proposes to use dissimilar metals either in contact or close enough for the gap between them to be bridged by an electrolyte, he shall satisfy the S.O. that adequate anti-corrosion provision is included in the design.

- (b) Structural Steel
 - (i) Rolled structural steel sections shall be mild steel Grade 43 or high yield steel Grade 50, both conforming to BS EN 10113 and BS EN 10210 and subject to the additional requirements of BS EN 1993-1-10. The dimensions, tolerances and properties of the structural sections shall conform to BS EN 10034 or BS EN 10056, BS EN 10067 or BS EN 10210.
 - (ii) Other structural metalwork shall comply with the requirements of relevant Singapore or British Standards.
 - (iii) Where the use of proprietary designs of prefabricated building frames is proposed, the standards to which they are manufactured shall be no less rigorous than those stated above.
- (c) Bolts and Nuts
 - (i) Steel bolts and nuts shall be high strength friction grip bolts conforming to BS EN 14399-1, Grade 4.6 bolts conforming to BS 4190, or Grade 8.8 or 10.9 bolts conforming to BS 3692, but with dimensions, tolerance and thread length to BS 4190, as detailed on the Drawings. Washers shall conform to BS 4320. Black metal washers to Form E shall be used unless otherwise specified.
 - (ii) High strength friction grip bolts shall be used in conjunction with approved proprietary load indicating washers.
- (d) Welding Consumables
 - (i) All welding consumables for welding structural steelwork (electrodes, wire, filler rods, flux, shielding gas and the like) shall comply with the requirements of BS EN 1011.
 - (ii) Welding of structural aluminium, where permitted by the S.O., shall be in accordance with Section 7 of the Aluminium Association's Specification for Aluminium Structures and/or the American Welding Society D1.2-83 Structural Welding Code for Aluminium.
 - (iii) In addition, weld electrodes for metal arc welding shall conform with BS EN ISO 2560 and to the requirements of the appropriate weld procedure.
- (e) Steel Items - General
 - (i) Steel items for which there is no relevant standard shall be fabricated as shown on the Drawings from steel which is in accordance with BS EN 10025, Grade 43 or Grade 50, by welding using procedures in accordance with BS EN 1011.
 - (ii) Unless otherwise specified or detailed on the Drawings, all steel items shall be painted in accordance with the general specifications.
- (f) Aluminium Items - General
 - (i) Items of aluminium and aluminium alloy shall comply with the relevant provisions of BS EN 485, BS EN 573, BS EN 603, BS EN 754 and BS EN 755. All aluminium and aluminium alloy shall be anodised. Structural aluminium shall be designed in accordance with BS EN 1999-1-1:2007+A2.

G1.4.5 Workmanship

(a) Design

- (i) Where design of structural steelwork is to be carried out by the Contractor, it shall comply with BS EN 1993-1-10. Where design of structural aluminium is to be carried out by the Contractor it shall comply with BS EN 1999-1-1:2007+A2. Designs shall be carried out and endorsed by a Professional Engineer (Civil or Structural) and shall be submitted to the S.O. for approval prior to commencement of fabrication. Where required, the Contractor shall be responsible for all Authorities submissions for approval.

(b) Detailing

- (i) The detailing and fabrication of structural metalwork shall be such as to minimise the need for site welding. Structures and components such as trusses, combined columns and beams shall be shop fabricated so as to form sub-assemblies of the largest practical size suitable for transportation, handling and erection. Connections between members at the erection stage shall generally be bolted.
- (ii) Structural metalwork shall be so detailed and fabricated as to minimise the formation of pockets to hold condensation, water or dirt. Where such pockets are unavoidable, suitable drainage shall be provided.
- (iii) Detailing shall be in accordance with BS EN 1993-1-10 and BS EN 1999-1-1:2007+A2 or other relevant standard.

(c) Detail Drawings and Welding Procedure Sheets

- (i) The Contractor shall prepare detailed shop and erection drawings and welding procedure sheets for all structural metalwork in accordance with BS EN ISO 15607 and these shall be submitted to the S.O. for approval before fabrication commences. The sequence of submissions shall match the order of fabrication and erection.
- (ii) Shop drawings for fabrication shall show full details of the type of steel, member sizes and dimensions, weld details, welding sequences and any requirements for weld stress relieving.
- (iii) Welding procedure sheets shall show all details of the welds including:
 - 1) Welding method;
 - 2) Current type, voltage and amperage;
 - 3) Welding positions;
 - 4) Preparation angles and methods;
 - 5) Number of runs and their welding positions;
 - 6) Type, size and class of electrode;
 - 7) Shielding gas type and flow rate (where applicable);
 - 8) Non-destructive testing methods and extent of coverage; and
 - 9) Pre-heat, post-heat and stress relief methods, temperatures and the like.

- (iv) Where substitution of alternative structural sections or modifications to the design or details is proposed, the Contractor shall prepare revised drawings, and supporting calculations if required, for consideration by the S.O.. Any such proposals by the Contractor shall not be put into effect unless the S.O. has given his approval.
- (v) Erection drawings shall detail all connections, showing bolt types, tensions and sizes and site weld details as applicable.
- (vi) Shop and erection drawings shall show all marks necessary to facilitate erection and the marks shown on the drawings shall be identical with those shown on the fabrications.

G1.4.6 Welding

- (a) All welding carried out during fabrication or erection shall be in accordance with the requirements of the relevant reference standards and as shown on the approved detail drawings.
- (b) Welding shall be a metal arc process in accordance with BS EN 1011, unless otherwise permitted by the S.O.. Details of the proposed weld procedures shall be submitted to the S.O. for approval at the same time as the detail drawings. All connections shall be welded in such a manner as to make the finished connections neat and smooth in appearance and suitable for painting. All slag shall be removed and any sharp projections shall be ground smooth.
 - (i) Before welding of structural steelwork is commenced either in the fabrication shop or on site, weld procedure tests shall be carried out in accordance with BS EN ISO 15607 where directed by the S.O..
 - (ii) Equivalent weld procedure tests shall be carried out for other structural metalwork.
 - (iii) A full time experienced third-party AWS Certified Welding Inspector (CWI) for the inspection of all field welding.
 - (iv) All welders for structural steelwork employed either in the fabrication shop or on site shall pass qualification tests relevant to the weld procedures in use in accordance with BS EN ISO 9606-1. In the case of welders engaged on fillet welding only, BS 4872 Part 1 is an acceptable alternative. Welders for other structural metalwork shall pass equivalent qualification tests. Welders shall have satisfactory evidence of having been engaged in welding for at least 9 months in the preceding 12-month period. If the work of any welders employed on the Contract is unsatisfactory, the Contractor shall carry out such further welder qualification tests as are necessary to demonstrate that the welders are proficient.
 - (v) All parts to be welded shall have scale, slag, loose rust, paint and other foreign matter removed by means of a mechanical brush and shall be left clean and dry immediately before welding. All scale and slag shall be removed from each weld run when it is completed before welding over any previously deposited weld metal and prior to air testing.

- (vi) The electric generators used for welding shall be specifically manufactured for welding and shall have such controls and regulators as are necessary to adjust and control the voltage and current of the arc so that the conditions under which the electrode works shall conform to those recommended by the makers of the electrode. Electrodes shall be supplied as necessary to suit the different conditions of working. All equipment shall conform to the regulations of the Relevant Authority or of any other authority recognised by the S.O..
- (vii) The Contractor shall comply with the SS CP 50:1997 "Code of Practice for Safety in Welding and Cutting (and Other Operations Involving the Use of Heat)" issued by the Ministry of Manpower. The Contractor is required to adopt the stipulated colour coding of gas hoses and to ensure that welding and cutting operations at the worksite conform to the Code of Practice.
- (viii) The Contractor shall note that no welding is allowed at any location where the existing gas transmission pipe is exposed.
- (ix) All gas cutting sets shall be fitted with hose check valves. These are non-return valves fitted to the blowpipe ends of the oxygen and fuel hoses to arrest the flame at the inlet connections of the blowpipe, should a flashback occur during welding or cutting operations. All welding tools are to be kept free from oil and grease and main cylinder valves shall be shut-off when carrying out maintenance works on welding equipment.
- (x) Welds shall be subjected to non-destructive testing by processes which may include but shall not necessarily be limited to radiographic, ultrasonic, magnetic particle, or dye penetration methods, depending on the type of weld and its position in the structure. The standards of acceptance for welding of structural steelwork shall be as defined in PD 5500 + A3, unless otherwise agreed with the S.O.. Equivalent standards shall apply for welding of other structural metalwork. In case there is a conflict the higher frequency level of non-destructive testing (NDT) shall apply:
 - 1) Butt Weld Joint: 100 percent radiographically tested and repaired;
 - 2) Penetration Welds: 100 percent ultrasonically tested and repaired;
 - 3) Fillet Welds: 100 percent examined and repaired, employing the magnetic particle inspection method using the half-wave rectified direct and alternating current; and
 - 4) All Welds: 100 percent visually inspected.
- (xi) If any work shows defects or fails to comply with the requirements of the Drawings or Specification for any reason it shall be repaired or rejected, even though it may have been carried out by qualified welders using approved procedures.

G1.4.7 Fabrication Tolerances

- (a) The general tolerance on all dimensions shall be in accordance with BS EN 1993-1-10. Holes shall be aligned such that fasteners can be freely inserted through the members at right angles to the contact face. Where holes in members cannot be aligned without damaging or distorting the structure or reaming the holes (where permitted by the S.O.), the member or members shall be rejected.
- (b) A structural member shall not deviate from straightness (or from the specified shape) by more than 1/1000 of the length between lateral restraints or 3 mm whichever is greater in the case of compression members and beams.
- (c) A structural member shall not deviate from its intended length by more than 3mm.
- (d) Lengths of components shall be such that cumulative variations do not prejudice the accurate alignment of the completed structure.
- (e) Where two metal surfaces are required to be in contact to affect a bearing or frictional contact, the surfaces shall be prepared so that at least 90% of the area is touching before any clamping force is applied.
- (f) Metalwork at floor level (e.g. joints between flooring sections and between steel flooring and adjacent concrete):

Difference in level at joints:	3	mm
Joint gap:	3	mm
Dimensions shown on drawings:	± 5	mm
Level shown on drawings:	± 5	mm

G1.4.8 Dissimilar Metals

- (a) Where dissimilar metals are used in proximity to structural members or their connections, contact between such metals shall be avoided unless the Contractor can demonstrate to the satisfaction of the S.O. that contact between the dissimilar metals will not lead to galvanic corrosion.
- (b) Contact between aluminium or aluminium alloy and galvanised mild steel will be permitted. For fixing aluminium to steel structures all fastenings such as bolts, nuts, washers and screws shall be zinc or cadmium plated.
- (c) Where zinc plated parts might otherwise become sacrificial anodes to the main structure, or where the electrolytic potential difference exceeds 250 mV, the parts shall be separated by an insulating medium of adequate strength.

G1.4.9 Identification of Members

- (a) All members of assemblies shall be marked at the factory with distinguishing numbers, letters or marks corresponding to those on approved drawings or parts lists. Such marks if impressed before painting shall be clearly readable afterwards.
- (b) Any temporary bolts for field erection shall be clearly marked as such to distinguish them from bolts for permanent connections.

G1.4.10 Storage

- (a) All structural elements and other materials shall be stored (whether at the place of fabrication or on Site) clear of the ground and shall be protected against deterioration from any cause. The Contractor shall agree to the storage areas on the Site with the S.O. before any structural steelwork is delivered.

G1.4.11 Erection

- (a) All erection processes shall be carried out in accordance with the recommendations of BS 5531 or other relevant standards. The equipment, machinery, apparatuses and tools used for handling and erection shall be suitable and safe. The Contractor shall supply, maintain and afterwards remove temporary protection to workmen and materials below any structural metalwork under erection and shall provide walkways and ladders so that inspection of any part of the completed structure may be safely carried out.
- (b) Anchor bolts shall be properly located by the use of templates or other approved method. After building in the anchor bolts, a sufficient time shall be allowed for concrete strength to develop before finally pulling down to shims and grouting.
- (c) Every part of the structural metalwork shall be positioned accurately in accordance with the dimensions on the approved drawings with a maximum tolerance of 5 mm except where the metalwork supports metal flooring, when the finished tolerances shall be such that differences in level between adjacent flooring sections or between flooring sections and adjacent building floors do not exceed 3 mm. Gaps between flooring sections shall not exceed 3 mm.
- (d) Where structural metalwork is to be grouted on or against a concrete or other similar surface and the design requires a uniform contact area under the base plate, or where directed by the S.O., the Contractor shall remove all shims or wedges and replace these with grout of the same constituents and consistency as the main grout. Such shims shall be initially positioned so as to aid removal. In the case of baseplates with large overhangs or with heavy bearing loads the positions of shims shall be agreed with the S.O. before the structure is erected on them.
- (e) During erection, the work shall be securely bolted or otherwise fastened, and if necessary temporarily braced, so as to make adequate provision for all erection stresses and conditions, including those due to erection equipment and its operation. No permanent bolting shall be done until proper alignment has been achieved.
- (f) The Contractor shall immediately report to the S.O. any errors in shop fabrication, and any deformations resulting from handling or transportation, which prevent proper assembly and fitting of parts. Methods to be used to correct such errors or deformations shall be agreed with the S.O. prior to beginning of any corrections.
- (g) The Contractor may use drift pins to bring several parts together but shall not use them in such a way as to cause distortion or damage.

G1.4.12 Installation

- (a) Due care should be taken in designing the fastening arrangements so as to avoid projected bolt heads above the finished surface levels.
- (b) Where metalwork is to be founded on and/or fastened to concrete the Contractor shall use one of the following methods:
 - (i) Bolting or screwing the metalwork to plates or angle sections having anchors cast into the concrete structure or set into pockets left in the concrete structure;
 - (ii) Setting the metalwork into pockets left in the concrete structure;
 - (iii) Bolting the metalwork to anchor bolts (studs) cast into the concrete structure or set into pockets left in the concrete structure;
 - (iv) Bolting the metalwork to anchor bolts (studs) as specified, in holes drilled into concrete structure; and
 - (v) Bedding the metalwork on cement mortar placed on the concrete surface.
- (c) Cement mortar of 3 parts of sand to 1 part of cement shall be used for bedding access covers and the like and for filling around metalwork or bolts set into pockets smaller than 50 mm square or equivalent. For larger holes, concrete of the same grade as the structure shall be used instead of mortar. The installation methods permitted for various types of metalwork are specified in the table below.

Items of installation	Metalwork permissible Methods
Stairs, ladders, flooring	(i), (iii) and (iv)
Step irons, sockets for removable items	(ii)
Handrailing	(ii), (iii) and (iv)
Access covers, surface boxes and gully gratings	(ii) and (v)

- (d) During installation, each item of metalwork shall be temporarily braced as necessary to resist all forces which are likely to be applied to it during installation, fixing and building in. Any bolted connections required as part of the installation operation shall be fitted and tightened before fixing bolts are tightened or pockets are grouted. The nuts of grouted or resin fixed bolts shall not be tightened until the grout or resin has fully cured.
- (e) Where approved by the S.O., small or lightly loaded items may be fixed using screws and plugs set in drilled holes.

G1.4.13 Bolted Connections

- (a) When assembled, all joint surfaces, including those adjacent to bolt heads, nuts and washers, shall be free from burrs, dirt or other deleterious matter or defects preventing proper seating of the parts.

- (b) For structural steelwork, the Contractor shall make bolted connections in accordance with the details shown on the shop or erection drawings and with BS EN 1993-1-10. The bolts used shall be in accordance with general specifications G4 and as detailed on the Drawings. When the bolt head or nut is in contact with a surface which is inclined at more than 30° from a plane at right angles to the bolt axis, a taper washer shall be placed to achieve satisfactory bearing. The bolt length shall be such that each bolt shall project through the nut not less than one thread plus the thread run-out and not more than five threads when fully tightened; the length of the threaded portion shall be such that there shall be at least two threads under the nut after tightening.
- (c) Additionally, foundation bolts shall have a minimum thread length equal to the bolt diameter plus 100 mm.
- (d) The Contractor shall ensure that the facing surfaces (i.e. those surfaces required to transmit friction in conjunction with high strength friction grip bolts) are kept free of contamination by grease, rust, moisture, dust and the like. If the surfaces are primed after fabrication, the primer shall be suitable for use in such connections and shall be protected from damage during transit or erection. Should contamination occur, it shall be remedied by appropriate means, such as:
 - (i) On coated surfaces, with a degreasing agent followed by washing and drying; and
 - (ii) On uncoated surfaces, as above, or by blast cleaning or wire brushing to clean steel.
- (e) These remedial measures shall be taken immediately before the surfaces are bolted together.
- (f) High strength friction grip bolts shall be used in accordance with BS EN 1993-1-8, using approved load-indicating washers to check bolt tension. They shall be tightened in accordance with the manufacturers' recommendations and the tension shall be rechecked not less than three hours after first tightening. The bolts shall then be re-tightened to the initial load, all to the approval of the S.O..
- (g) For other structural metalwork bolted connections shall be formed in accordance with relevant standards.
- (h) Under no circumstances shall predrilled holes be altered on site to accommodate out of alignment. In particular, burning of predrilled holes using oxy acetylene torches shall not be carried out. Such activities once brought to S.O.'s attention will result in removal of the affected parts from Site.
- (i) Bolts for fixing metalwork to concrete, brickwork or blockwork shall be of the following types:
 - (i) For fixing in drilled holes in horizontal, vertical or inclined surfaces, approved proprietary epoxy resin fixed bolts shall be used. The epoxy resin shall be a quick setting formulation, supplied in suitable premeasured sachets to separate the components until punctured in the hole and mixed by rotation of the fixing bolt. The resin and bolts shall be used in accordance with the manufacturers' instructions. When fixed, the resin shall completely fill the space between the bolt and the hole to the surface of the concrete, brickwork or blockwork;

- (ii) As an alternative to (i), approved mechanical expansion type bolts may be used for fixing to concrete surfaces inside buildings above ground level where the application proposed by the Contractor is approved by the S.O.. Expansion bolts shall not be used inside liquid retaining structures, in chambers, basements and similar locations below ground level, in exterior locations, or in any brickwork or blockwork;
- (iii) For fixing to horizontal top surfaces of concrete by grouting with cement based grout or mortar, foundation bolts may be used in preformed, drilled or cut pockets; and
- (iv) Where approved by the S.O., screws and approved fixing plugs may be used.
- (j) The Contractor shall submit details of his proposed fixings for the approval of the S.O.. Where required by the S.O., the Contractor shall demonstrate the effectiveness of the fixings by suitable tests.
- (k) Other bolts and nuts used in fabrication or erection shall be in accordance with BS 4190 or BS 3692.
- (l) Bolts, nuts, screws, washers and other fixings shall be stainless, galvanised, sheradised or electro plated steel in accordance with the provisions of the general specifications.

G1.4.14 Ladders

- (a) Unless otherwise specified, ladders shall be constructed of hot dip galvanised structural steel, ladders shall be constructed to BS EN 10025 Grade 43.
- (b) Where aluminium alloy been specified, it shall be to BS EN 573, BS EN 603 and BS EN 755. If stainless steel been specified, ladders shall be constructed to PD 970 Grade SS316.
- (c) Rungs shall be of substantial section serrated tube or surface profiled to enable sure hand and foot holds.
- (d) The ladders shall be of all welded construction and shall comply with the requirements of BS EN 131-1:2007+A1:2011 for the spacing of stringers, rungs, safety hoops and hand holds. Ladders of all types shall be fixed with stainless steel bolts, nuts and washers to PD 970.
- (e) Unless otherwise specified or shown on the Drawings, ladders shall have the following dimensions:

Minimum thickness of stringer:	12	mm
Diameter of rungs:	20	mm
Rung spacing:	250	mm
Rung length between stringers:	400	mm
Lower most rung to finished floor:	175	mm
Minimum clearance to wall or other obstruction:	20	mm
Maximum distance between stringer supports:	200	mm

- (f) Wherever possible, stringers shall extend 1.0 m above the level of the floor or platform at the top of the ladder and the clear width between stringers shall be increased to 600 mm for this length.
- (g) The stringers shall be sized to suit the height of ladder and the intervals of the stringer supports. Stringers shall be drilled to receive the rungs, which shall be welded to the stringers on each side of each stringer. For the bottom ends of the stringers design not for floor fixing, it shall terminate at wall fixing supports at least 150 mm above the floor and design and endorsed by P.E and approval by S.O.. All edges of stringers, brackets, slice plates and other components shall be ground smooth to remove burrs and sharp edges.
- (h) Ladders exceeding 3 m high shall be provided with safety hoops at intervals not exceeding 900 mm, with the lowest hoop 2.5 m above the ladder foot unless otherwise specified or shown on the Drawings. Unless provided otherwise, landings of open mesh flooring and handrails as specified shall be provided at intervals not greater than 9 m. When detailed on the Drawings, anchorage points shall be provided for attaching safety harnesses instead of the above safety hoops.
- (i) All access ladders that connect to parapet walls/safety railings/barriers at the top shall be provided with a swing gate at the top of the ladder. The swing gates at the top of access ladders shall not be locked thus it shall be readily openable for exit access. Access ladders and other means of access shall be designed and provisioned to comply with applicable codes of practices and prevailing statutory requirements, i.e. WSH Code of Practice, BCA requirements and approved documents. The swing gates shall be constructed with the same materials and finishes as the access ladders attached to it, the gates shall be designed as 'safety barrier' as part of safety railings/barrier/parapet walls, to comply with BCA requirements and WSH requirements. Ladders designated as exit access shall comply with the applicable fire safety requirements.
- (j) Step Irons: Step irons shall be bent from 304 stainless steel or extruded BS H9 aluminium alloy tubes 20 mm diameter x 3 mm thick inserted by machine in plugs. The fixing plugs of the step iron shall be manufactured from high density linear polyethylene material that complies with BS 6076.

G1.4.15 Walkways and Access Platforms

- (a) Standard structural steel sections shall be used for the support structure.
- (b) Toe plates shall be fitted along the outer edges of all walkways and shall be part of the structure and not the floor panels. Toe plates shall extend 100 mm above the top level of the floor panels. Floor panels shall be sized so that each panel does not weigh more than 50 kg.
- (c) Provisions shall be made in the design for adjustment to eliminate irregularities in structural floor levels.
- (d) Where dissimilar materials come into contact with each other an insulating membrane or paint coating shall be applied to minimise direct contact.
- (e) All components shall be hot dip galvanised after fabrication in accordance with BS EN ISO 1461.

- (f) Unless otherwise specified, metal surfaces shall be coated with two (2) layers of anti-rust galvanizing paint to BSEN ISO 1461:1999 and (2) layers of epoxy coating to BS 6920: Part 3 on the interior and exterior of the steel surfaces.
- (g) Walkways and flooring shall be designed to support uniformly distributed loads of not less than 2 kN/m² and for a concentrated load of not less than 1.5 kN over any square area of side 200 mm whichever produces the greater stresses or deflections and of sufficient section to withstand the design and working loads without deflecting more than 5 mm. The loadings shall be considered a minimum requirement and if an item of plant demands larger loading the loadings shall be increased accordingly.
- (h) Unless otherwise specified, platform, railings I-beams, supports, etc. shall allow for at least two (2) medium-sized personnel to walk and work safely. All supports shall be designed to ensure that the platform and railings are stable and safe for minimum live load of 250kN/m² in addition to the dead load as required and shall be design and endorsed by the P.E, all P.E. endorsed calculations and design shall submitted to S.O. for approval.

G1.4.16 Access Stairways

- (a) The staircases and walkways are to be formed by the bolting of structural steel members to the concrete, no supporting members may be built in. The use of stanchions will not be permitted unless it can be shown that support by other means is impracticable.
- (b) Stair treads shall be purpose made with a rounded front nosing and rear end side trimming and shall be anti-slip. The treads shall be firmly bolted in position and the arrangement, width, height, and going of treads shall be conducive to safety, not inferior to the proposed arrangement on the Contract drawings and subjected to the detailed approval of the S.O..
- (c) Stairways shall be designed for a loading of 5.0 kN/m² of plan area of the stairway. Stairways shall be provided with handrailing and stringers of cross section suitable for the span and loading. Treads shall be of open bar gratings, expanded aluminium grating panels or cold-formed planks as shown on the drawings. Except where so shown on the drawings, treads of pressed bar construction shall not be used.
- (d) The angle of rise of stairways shall be between 30° and 42°. The rise between treads shall be uniform and between 155 mm and 190 mm. Stairways in the same area of the works and in similar locations shall have the same angle of rise and the same rise between treads.
- (e) The depth of the treads shall be between 250 mm and 300 mm. The width of the stairways shall be as shown on the drawings. The bottom tread shall not be more than 175 mm from finished floor level.
- (f) The stringers shall be mounted by means of angle brackets with slotted holes for adjustment of line and level.
- (g) Stairways shall have 100 mm toe boards in aluminium, which shall be securely attached to the handrail standards or structural members.
- (h) Flooring panel's provision shall be made to allow free drainage.

- (i) All structural steel and its associated components shall be hot dip galvanised after fabrication in accordance with BS EN ISO 1461.

G1.4.17 Access Covers, Road Gully Gratings and Frames

- (a) Except where otherwise detailed on the Drawings, access covers, road gully gratings and frames shall be of ductile iron and shall comply with BS EN 124. Lockable units shall be provided for the locations specified or as shown on the Drawings.
- (b) Covers and frames detailed as sealed shall incorporate suitable neoprene or other approved synthetic rubber sealing rings or shall seal by another approved sealing method.
- (c) Gratings to drainage channels shall be provided where the channel is within a building, where there is regular pedestrian access such as access to a door or equipment, where the depth of the drain exceeds 600mm and also where shown on the drawings.
- (d) Manhole covers and frames for concrete manholes in roads, verges, driveways, etc. which may be subject to traffic loadings shall be grade 'A' (heavy duty), ductile iron, to BS EN 124 and shall be of the three-point suspension, non-rocking type. They shall be carefully set to the slope of the ground or road surface. Except in roads and other paved areas the covers shall be set 150 mm above ground level, and neatly surrounded by concrete haunching.
- (e) The covers and frames for concrete and brickwork manholes in open fields, residential properties, etc. which would not normally be subject to traffic loadings, shall be Grade 'B' (medium duty), ductile iron to BS EN 124. They shall be carefully set to the slope of the ground or road surface. Except in roads and other paved areas the covers shall be set 150 mm above ground level, and neatly surrounded by concrete haunching.
- (f) The covers and frames for concrete and brickwork manholes in open fields, residential properties, etc. which would not normally be subject to traffic loadings, shall be Grade 'B', ductile iron to BS EN 124.
- (g) Manhole covers on all sewers shall be embossed with the letters "SEW". After setting in position, the lifting and prising holes provided in the cover shall be cleaned out and filled with tarred hemp.
- (h) Fall protection grating system shall be provided below the hatches and cover, it shall be visible and lock automatically in the full open position, and provision for locking to prevent unauthorized opening. Type 316 stainless steel hold open device shall be provided to lock the cover in the fully open 90 degree position. All hardware shall be Type 316 stainless steel.

G1.4.18 Surface Boxes

- (a) Surface boxes for key operation of valves and sluice gates shall be of grey or ductile iron and shall comply with the relevant requirements of BS 5834: Part 2. Except where otherwise specified or shown on the Drawings, they shall be Grade A and coated.

G1.4.19 Chequer Plate and Open Mesh Metal Flooring

- (a) Metal mesh flooring shall be constructed in hot dip galvanised steel unless otherwise specified.
- (b) All marine grade Aluminium members shall be anodised after fabrication.
- (c) Flooring and walkway supports shall be independent of rotating machinery bearing supports to prevent the setting up of oscillations and noisy vibrations.
- (d) Structural members shall be of appropriate sufficient section to withstand the design and working loads without excessive deflection.
- (e) Flooring, and walkways shall be designed to support uniformly distributed loads of 5kN/m² and for a concentrated load of 5 kN over any square area of side 300 mm whichever produces the greater stresses or deflections and of sufficient section to withstand the design and working loads without deflecting more than 5 mm. The loadings shall be considered a minimum requirement and if an item of plant demands larger loading the loadings shall be increased accordingly.
- (f) Open metal flooring shall be constructed in accordance with BS EN ISO 14122 Parts 1 to 3.
- (g) The individual ribs of the open mesh flooring shall be firmly welded or otherwise connected to each other at both top and bottom at every point of contact.
- (h) Chequer plates shall be aluminium alloy as minimum and shall have a non-slip pattern on its upper face and shall be securely bolted with countersunk screw studs to frames set in the edges of and spanning the duct or opening. The sizes of the chequer plate flooring shall be such that no discernible deflection occurs under the weight of one man standing upon it. Where it is not practicable to provide individual frames to each chequer plate, the plates shall be stiffened by welding structural aluminium angles to the underside to eliminate discernible deflection. Chequer plates for use in valve chambers and manholes shall be hinged to one side of the frames unless otherwise specified or shown on the Drawings.
- (i) Panels shall be so arranged and sized that they can easily be removed by two men without the removal of adjacent panels or equipment. Each panel shall not be heavier than 20 kg and shall be provided with suitably sized lifting holes.
- (j) The edges of all panels shall be trimmed and kerbed and gaps between units and the adjoining floors, etc. shall not be more than 5 mm.
- (k) The flooring panels shall be arranged to seat firmly as independent sections. No reliance for support may be placed on the interconnection or clipping of panels together. Clips provided to interconnect panels in line shall be of an approved design allowing for ready removal and not protruding above the adjoining surface levels.
- (l) Panels are to be provided with trimmed cut-out holes for lifting (not less than two per panel) and for fitting around pipework, headstock spindles, etc.
- (m) The edges of all flooring, platforms, etc. shall be finished with approved kerbs. The kerbing shall be provided by the flooring supplier and shall be fitted with fish tail lugs for building into the concrete work.

G1.4.20 Handrailing

- (a) The Contractor shall design, supply and install handrailing and all members necessary for their support as shown on the drawings unless otherwise specified.
- (b) All handrail shall be marine grade aluminium alloy as minimum unless specified otherwise. Handrailing shall be anodised aluminium alloy to BS 1474. The rails, standards etc. shall have an external diameter of not less than 38mm. Handrails shall be designed to support the horizontal loads to comply with Table 4, BS6399 Part 1.
- (c) To avoid electrolytic action, where aluminium surfaces are in contact with concrete surfaces or with dissimilar metals, rubber or PVC gaskets shall be used to isolate the aluminium and no separate measurement will be made for the latter.
- (d) All aluminium members shall be anodised and sealed against corrosion after fabrication.
- (e) Aluminium handrailing shall be anodised aluminium alloy standard handrailing and comprise double line rails made from 44 mm outside diameter tube with 4 mm wall thickness and comply with BS EN 573, BS EN 603 and BS EN 755.
- (f) Handrails shall be curved to smooth radii at bends and corners and all joints except for removable sections, shall be made to standards. The external finish shall be completely smooth, free of all blemishes and rough areas.
- (g) The handrail rails shall be at heights of 1000 mm and 500 mm. Ball joints at stanchion/rail junctions are not acceptable. The rails and stanchions shall each be constant section, where toe plates are specified they shall be a minimum of 5 mm thick and shall be not less than 100 mm in depth measured from the floor level. Base fixing shall be stainless steel with nuts and washers. Where removable lengths of handrails are detailed, half lap joints shall be used, all fixing accessories shall be stainless steel.
- (h) Safety chains shall be stainless steel to PD 970 fixed at one end to a loop and at the other end to a hook. The loop and hook shall be of the same material as the standard.
- (i) Removable bases shall be furnished with a dowel which shall fit into a socket cast into or bolted onto the concrete or metal platform. The socket shall be of the same material as the standard.

G1.4.21 Bollards

- (a) Bollards shall be mild steel Grade S275 unless specified otherwise. Paint colours to be agreed with S.O..

G1.4.22 Servicing Hooks

- (a) Servicing hooks shall comply with those provisions of BS EN 1677 which relate to the appropriate type. Each hook, together with any beams and other supporting members, shall be designed (with an appropriate allowance for dynamic loading) to carry safely the heaviest item of plant required to be lifted.
- (b) The safe working load shall be clearly and permanently displayed alongside each hook so as to be legible from the normal working level.

G1.4.23 Cover Keys and Opening Tools

- (a) A minimum of two pairs of keys for use with each type of cover shall be handed over by the Contractor at the completion of the Contract. Such keys shall be those recommended by the manufacturer of the cover.
- (b) Lifting keys and devices for unfastening locking screws shall be of galvanised mild steel. They shall be supplied at the rate of one set for each five similar covers, with a minimum of two sets of each particular type.

G1.4.24 Electrolytic Action

- (a) Where aluminium surfaces are attached to or in contact with concrete surfaces suitable PVC washers and packings shall be used to isolate the metal. Where large areas of contact are arranged the concrete surface may be treated with bituminous paint. Suitable measures shall be taken throughout the erection of the metalwork to avoid any metal to metal contacts likely to induce electrolytic action and corrosion.

G1.4.25 Access Covers to Covered Process Units

- (a) The access covers to covered process units shall be aluminium hinged type. They shall be designed for a pedestrian load of a minimum of 5.0 kN/m² and the proposed design shall be approved by S.O. prior to construction.
- (b) Access covers shall be designed for opening and closing by one person.
- (c) The frame of the covers shall be embedded into the concrete structure. The clear opening size of the cover shall be as shown on the Drawings.
- (d) All fittings for the covers such as handles, hinges, levers and screws shall be stainless steel. The covers shall be designed and constructed to be opened and closed with ease, and when closed shall be completely air tight with necessary sealant gaskets to prevent entry of mosquitoes.

G1.4.26 Testing

- (a) The S.O. may inspect structural metalwork during fabrication and witness testing at the fabrication works. The Contractor shall give the S.O. adequate notice of such operations in order that the necessary arrangements may be made.
- (b) The Contractor shall arrange for the S.O. and his authorised representatives to have access to the fabrication works at all reasonable times and to be provided with all the necessary facilities and test equipment to carry out such inspections.
- (c) Before any steel overhead runway beam is taken into use for the first time, the Contractor shall submit to the S.O. a signed certificate from the manufacturer in accordance with BS EN 1993-6.

END OF SECTION